

ACME: Adaptive Cognitive Memory Engine

Externalizing Belief, Memory, and Learning from LLM Weights

A cognitive memory substrate for external belief lifecycle management in LLM agents.

Mohamed Kamil Bourouiba

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LLM □ Episodic Memory □ Belief Engine □ Feedback Loop □ Updated Beliefs

Code: <https://github.com/KamilBourouiba/ACME> (tag v0.1.1-arxiv)

Project site: <https://kamilbourouiba.github.io/ACME/>

Data: GET /api/v1/benchmark/export (Postgres benchmark_runs)

Abstract

Large language models can retrieve context, but they do not natively maintain durable episodic memory, explicit belief states, contradiction-aware revision, or feedback-driven correction. We present **ACME** (Adaptive Cognitive Memory Engine), an external cognitive memory substrate that treats the LLM as a language processor while delegating persistence, confidence tracking, belief promotion/demotion, and outcome-based correction to specialized services. ACME introduces a belief lifecycle from Observation to Inference, Hypothesis, and Belief, governed by a Cognitive Reliability Score (CRS). The system combines graph memory (Neo4j), vector retrieval (pgvector), and optional contrarian verification. We evaluate ACME on **MemoryBench v3.1**, a sandbox-isolated benchmark for retention, groundedness, feedback correction, and belief quality, and on the official **LongMemEval** oracle split (500 Q). On GPT-4.1, ACME reaches overall **0.925** on MemoryBench primarily because it is the only system scored on feedback and belief dimensions; retrieval-style baselines remain competitive on retention and groundedness (≥ 0.90) but score N/A on belief/feedback, capping their overall near 0.49. On LongMemEval, ACME reaches **87.6%** vs. **77.6%** (RAG-like baseline). These results suggest that explicit external belief management can complement retrieval and long-context modeling for long-term LLM agents.

1. Introduction

Retrieval-augmented generation (RAG) [8] augments LLMs with document search but does not maintain explicit epistemic states, handle contradictory evidence, or learn from prediction failures. LongMemEval [35] shows commercial assistants and long-context LLMs suffer **30–60%** accuracy drops on sustained interactive memory versus oracle evidence — retrieval alone is insufficient. Hindsight [48] argues current agent memory cannot distinguish *what was observed* from *what is believed*, nor maintain preference-consistent reasoning across sessions. Reflection-Bench [49] identifies **belief updating** and meta-reflection as core yet underdeveloped dimensions of epistemic agency in LLMs.

Dense and hybrid retrievers [13,14] improve recall yet remain stateless: they neither promote stable beliefs nor demote refuted hypotheses. Agent memory frameworks — MemGPT [9], generative agents [15], Zep [19], and LangGraph-style orchestration [10] — extend context windows via paging, temporal graphs, or state graphs but lack auditable belief lifecycles and structured feedback loops comparable to cognitive architectures [2,16].

Episodic and semantic memory have long been distinguished in psychology and AI [17,18]. Recent work on long-term conversational memory (MemGPT, memory streams [15], Zep [19]) focuses on *what to store and retrieve*, not *when accumulated evidence warrants belief*. ACME closes this gap with an explicit promotion pipeline (Observation -> Inference -> Hypothesis -> Belief) and symmetric demotion under contradiction — aligning with truth-maintenance and belief-revision traditions [20,21].

Contributions

1. A CRS-governed belief lifecycle with promotion and demotion under contradiction.
2. A hybrid graph + vector retrieval substrate (PostgreSQL/pgvector + Neo4j).
3. Contrarian verification for high-confidence answers.
4. MemoryBench v3.1 with scored belief and feedback metrics.
5. Reproducible Azure-hosted benchmark runs on GPT-4.1 and GPT-4.1-mini.

Thesis. RAG optimizes recall; ACME optimizes epistemic state—when experiences become beliefs, how contradictions demote them, and how outcomes close the prediction loop. Our claim is not universal SOTA on every long-context benchmark [35,36]. To our knowledge, ACME is among the first Azure-deployed systems to jointly evaluate **feedback correction**, **belief quality (CRS)**, and **contrarian groundedness** under per-scenario sandbox isolation against RAG-like, MemGPT-inspired, and LangGraph-style reference baselines [47].

2. Related Work

Retrieval-augmented generation. Lewis et al. [8] established the retrieve-then-generate paradigm for knowledge-intensive tasks. Subsequent systems add re-ranking [13], query rewriting [26], and self-critique — Self-RAG [27] and Corrective RAG [28] route or revise retrieval based on generation quality. ACME differs by persisting structured beliefs and failure traces outside the LLM, not only refining a single-turn retrieval pass.

Graph-augmented memory. Knowledge-graph RAG [29,30] and GraphRAG [29] extract communities and summaries from document graphs. ACME maintains a *live* typed graph (entities, causal edges, tenant scope) updated on every experience, coupled with vector recall over episodic embeddings — hybrid retrieval akin to [4,5] but with belief-weighted context assembly.

Agent memory systems. MemGPT [9] virtualizes context via OS-inspired paging; generative agents [15] use memory streams and reflection; LangGraph [10] composes stateful agent workflows. These systems improve *capacity* and *session continuity*; ACME adds CRS-governed belief promotion, contradiction logging, and prediction-outcome loops absent from reference baselines in our evaluation.

Learning from feedback. Reflexion [31] and ExpeL [32] use verbal reinforcement from trial outcomes; Constitutional AI [33] shapes behavior via principles. ACME’s feedback engine ties outcome signals to specific graph-backed beliefs, adjusting confidence and status (e.g., Challenged -> Deprecated) rather than only updating prompt-level reflections.

Cognitive architectures. ACT-R [2] and SOAR [16] model declarative memory, production rules, and learning from impasse. ACME is not a full cognitive architecture but borrows the separation of *fast reasoning* (LLM) from *slow accumulative memory* (Postgres, Neo4j, belief records) — similar in spirit to dual-process accounts [3] and complementary learning systems [34].

Evaluation of memory systems. Benchmarks such as LongMemEval [35], LoCoMo [36], MemoryBank [37], MemoryAgentBench [50], and MemBench [51] stress long-horizon recall, test-time learning, or reflective memory. Reflection-Bench [49] evaluates belief updating as one of seven epistemic dimensions. To our knowledge, no public benchmark in our comparison jointly scores sandbox-isolated **feedback correction**, **CRS belief quality**, and **contrarian groundedness** in a deployable API. MemoryBench v3.1 is designed to fill this gap (Table 1).

Belief-first and epistemic agent memory (current work). NeuSymMS [52] couples neural extraction with symbolic TMS via rule engines. Graph-native cognitive memory architectures [53] formalize versioned belief revision for auditability. Hindsight [48] achieves strong LongMemEval/LoCoMo scores via retain-recall-reflect tiers but does not expose CRS-governed promotion thresholds or prediction-outcome loops. **ACME** differentiates by (i) an integrated seven-engine cognitive loop deployed as a REST API, (ii) MemoryBench v3.1 scoring belief/feedback dimensions absent from retrieval-centric baselines, and (iii) persisted benchmark_runs for third-party reproduction.

3. System Design

Ingestion: LLM extraction [38] plus deterministic normalization -> Neo4j entities/relations + episodic embeddings [22].

Query: Hybrid retrieval (graph neighborhood [23,29] + pgvector cosine [4]) -> LLM reasoning -> optional contrarian pass [6,7] when confidence ≥ 0.8 .

Feedback: Outcome signals update beliefs [20], log failures [31], trigger consolidation (scheduled consolidation, 6 h).

CRS (Cognitive Reliability Score) = 40% prediction success + 20% temporal stability + 20% contradiction resistance + 20% source diversity — a composite reliability index analogous to calibrated confidence under heterogeneous evidence [39,40].

Causal edge types: observed_with, precedes, correlates, causes, disproves — distinguishing correlation from causation per Pearl and epistemic traditions [41,42].

Worked example (contradiction scenario). Scenario contradiction_handling ingests: “*Latency causes churn in enterprise segment.*” ACME extracts entities and promotes a causal belief with initial CRS. The user queries “*Does latency cause churn?*” with contrarian verification enabled. Feedback reports failed outcome; the belief engine marks the belief **Challenged**, lowers CRS, and logs the contradiction. A subsequent query no longer treats the link as high-confidence fact. This trace is persisted in Postgres/Neo4j and scored by MemoryBench’s feedback and belief-quality metrics—capabilities absent from vector-RAG baselines.

Multi-tenancy: Postgres rows and Neo4j nodes keyed by tenant_id (header X-Tenant-ID).

Ablation toggles: ABLATION_DISABLE_CONTRARIAN, ABLATION_DISABLE_BELIEF_SYNC, ABLATION_DISABLE_VECTOR (see Results).

4. MemoryBench v3.1

MemoryBench is an **internal, sandbox-isolated** evaluation suite (13 scenarios in the primary v3 compare; 14 with the v3.1 knowledge-update probe). Each scenario defines scripted episodes, a query, expected concepts, and optional feedback/contradiction hooks. Full scenario definitions live in acme/evaluation/memorybench.py; summary examples include retention_latency_churn (three latency episodes → “*Why do customers churn?*”), contradiction_handling (failed feedback + belief demotion), and knowledge_update (superseded Q1→Q3 pricing evidence).

Four metrics, LLM-as-judge [11,12] for retention and groundedness (keyword overlap reported separately in details.keyword_retention_avg):

Table 1: MemoryBench v3.1 metrics.

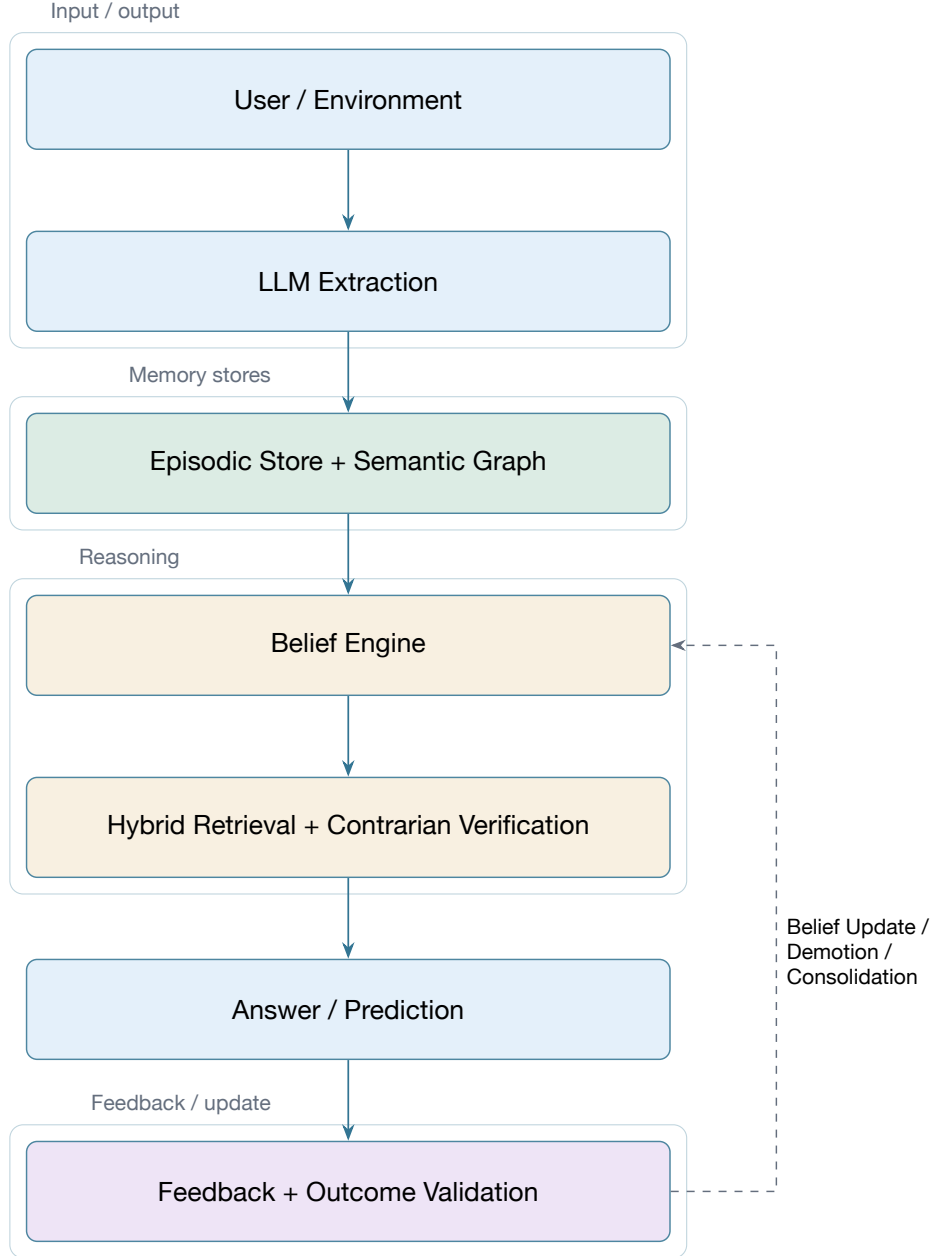


Figure 1: ACME cognitive loop: extraction, hybrid memory, belief governance, and closed-loop correction.

Metric	Description
Retention	Concept coverage (synonyms accepted)
Groundedness	Answer supported by ingested episodes
Feedback correction	Belief adjustment after contradictions
Belief quality	Mean CRS across tracked beliefs

Overall = unweighted mean of all four metrics. RAG-like, MemGPT-inspired, and LangGraph-style baselines do not maintain explicit belief records or

structured feedback traces; their Feedback and Belief Quality scores are marked **N/A** (excluded from their overall mean), which caps baseline overall near **0.48–0.49** even when retention and groundedness are strong.

4.1 Evaluation validity

Judge model. Retention and groundedness use the same Azure OpenAI deployment as the answer model (GPT-4.1 in primary runs; GPT-4.1-mini in sensitivity runs) via `evaluate_answer_quality` (`acme/llm/base.py`).

Blinding. The judge receives only the scenario

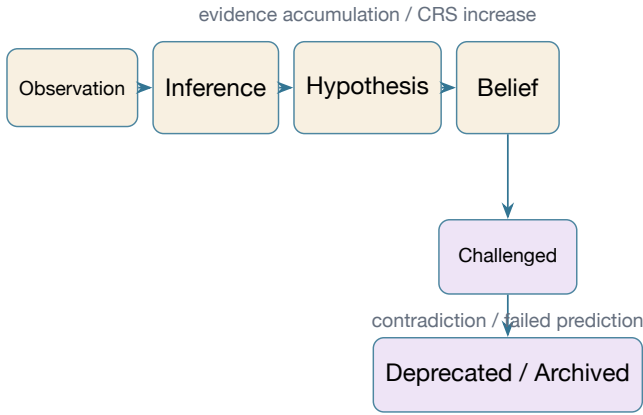


Figure 2: Belief lifecycle: promotion along evidence accumulation and demotion under contradiction.

question, system answer, reference concepts, and ingested episode text. It does **not** receive system identity, belief graph dumps, or CRS values—the same prompt template is used for ACME and all baselines.

Prompt rubric. The judge returns JSON with `retention_score` (semantic concept coverage; synonyms allowed) and `groundedness_score` (support by ingested episodes vs. hallucination). Temperature is **0.0**; on API failure a deterministic synonym fallback is used (`acme/evaluation/scoring.py`).

Non-judge metrics. Feedback correction is scored from persisted belief status changes and outcome hooks (`expect_belief_adjustment`, `contradiction_belief`). Belief quality is the mean CRS over active beliefs after the scenario—not LLM-judged.

Runs and variance. Reported jobs are **single end-to-end compare runs** per model tier (e.g. job 3b31e5e3, 725 s, zero scenario failures). We do not yet report confidence intervals; judge variance [11,12] remains a limitation (Section 7).

Sanity checks. Unit tests in `tests/test_memorybench.py` lock scenario structure and scoring plumbing; keyword-retention averages are exported alongside semantic judge scores for cross-checking.

Fourteen scenarios: retention, contradiction, multi-source conflict, error injection, **knowledge update** (LongMemEval KU [35]), long-term retention, hallucination resistance, feedback adjustment, adversarial noise, long-horizon noise, tenant isolation, healthcare domain transfer, multi-session recall, prediction-outcome loop.

Isolation: Each scenario deletes prior benchmark-tagged Postgres rows and Neo4j subgraph before run (`acme/evaluation/sandbox.py`) — preventing cross-scenario contamination noted as a failure mode in long-horizon evals [35,36,49].

Baselines. We compare against **minimal reproducible reference implementations**—not full upstream MemGPT or LangGraph products. Labels throughout: **RAG-like** (Lewis et al. retrieve-then-generate [8]), **MemGPT-inspired** (paging-style context buffer per Packer et al. [9]), and **LangGraph-style** (accumulated session state graph per LangChain docs [10]). See `docs/BASELINES.md`.

Primary reported results use the **13-scenario v3** suite (job 3b31e5e3, `pre-knowledge_update`) for strict comparability; v3.1 adds the LongMemEval-aligned knowledge-update probe.

Key finding. Among the benchmarks surveyed in Table 1, MemoryBench v3.1 is the only suite in our comparison that jointly scores **feedback correction**, **CRS belief quality**, and **contrarian groundedness** under per-scenario sandbox isolation.

5. LongMemEval (industry benchmark)

To complement MemoryBench, we integrate the official **LongMemEval** oracle split [35] (500 questions, evidence sessions only). Chat histories are ingested via the same production orchestrator; answers are scored with the **official LongMemEval yes/no judge prompts** from the reference implementation.

Protocol. Each question resets sandbox state (`longmemeval` tag). Haystack sessions are serialized as multi-turn user/assistant transcripts and ingested as experiences. ACME uses a **transcript-first** answer path: sessions are ranked newest-first, the LLM reads full chat transcripts (with belief graph as secondary context), and knowledge-update items trigger belief demotion on superseded sources. RAG-like and MemGPT-inspired baselines answer from retrieved context; an independent LLM judge (same deployment family as MemoryBench) applies type-specific rubrics (knowledge-update, temporal-reasoning, abstention, etc.).

Production run (June 2026): job 45623ca0, image `acme-api:longmemeval-v5-hybrid`, **500 Q clean**

Table 2: Benchmark positioning — evaluation capabilities (Table 1a).

Capability	LongMem Eval	MemAgent Bench	Reflection Bench	Mem Bench	Memory Bench
Long-horizon / multi-session	Yes	Yes	Partial	Yes	Yes
Knowledge update	Yes	Yes	Partial	Partial	Yes
Feedback / belief revision	No	Partial	Yes	Partial	Scored
Hallucination / abstention	Yes	Partial	Partial	Partial	Yes

Table 3: Benchmark positioning — infrastructure and belief metrics (Table 1b).

Property	LongMem Eval	MemAgent Bench	Reflection Bench	Mem Bench	Memory Bench
Per-scenario sandbox	No	Partial	Yes	Yes	Yes
Contrarian verification	No	No	Partial	No	Yes
Belief quality (CRS)	No	No	No	Partial	Yes
Production API + persisted runs	No	No	No	No	Yes

(~6.2 h). Summary: [benchmark-results/longmemeval-v5-full-500q.json](#).

Reproduction:

```
bash scripts/download_longmemeval.sh
bash scripts/run_longmemeval_prod.sh
python scripts/run_longmemeval.py \
  --types knowledge-update --systems acme,rag,memgpt
```

(Azure API for the prod script; full 500 Q: LONGMEMEVAL_TYPES=all.)

Results export to [benchmark-results/longmemeval-latest.json](#). LongMemEval measures **QA accuracy over chat history**; MemoryBench measures **belief lifecycle and feedback** — the two benchmarks are complementary and must not be merged into a single score.

Key finding. On the full LongMemEval oracle split (500 Q, GPT-4.1, single clean run), ACME v5 hybrid routing reaches **87.6%** overall vs. **77.6%** (RAG-like) and **78.6%** (MemGPT-inspired) — +10.0 points over RAG-like — with gains on temporal-reasoning (80.3%), abstention (83.3%), and knowledge-update (94.4%).

6. Experimental Setup

Table 7: Experimental configuration (Azure production).

Component	Configuration
LLM	Azure OpenAI GPT-4.1
Embeddings	text-embedding-3-small, 256D
Database	Postgres Flexible + pgvector
Graph	Neo4j 5.x on Container Apps
Judge	Same deployment as extractor/reasoner
Reproducibility	/benchmark/memorybench, async compare

Model-sensitivity runs use GPT-4.1-mini on the same Azure endpoint family. An additional run on a preview gpt-5.4 Azure deployment is reported in Appendix~C (not a publicly documented OpenAI product name at submission time).

Table 4: LongMemEval oracle protocol (industry benchmark [35]).

Component	Configuration
Dataset	longmemeval_oracle.json (500 Q, evidence sessions)
Ingest	ACME orchestrator / RAG-like / MemGPT-inspired on identical haystacks
Scoring	Official type-specific yes/no judge (reference evaluate_qa.py)
Metrics	Overall accuracy + per question_type breakdown
Reproduce	scripts/run_longmemeval.py (see docs/LONGMEMEVAL.md)

Table 5: LongMemEval knowledge-update subset (78 Q, oracle, GPT-4.1, June 2026).

System	Overall	KU	Abst.
ACME (transcript-first)	0.897	0.931	0.500
MemGPT-insp.	0.872	0.903	0.500
RAG-like	0.859	0.875	0.667

Table 6: LongMemEval oracle full split (500 Q, GPT-4.1, June 2026).

Sys.	All	KU	Multi	Temp.	SS-U	SS-A	SS-P
ACME	0.876	0.944	0.793	0.803	1.000	1.000	0.900
MemGPT-insp.	0.786	0.861	0.793	0.630	1.000	0.982	0.600
RAG-like	0.776	0.875	0.793	0.622	1.000	0.964	0.467

7. Results

Run metadata: API revision `acme-api--membenchfix`, compare job `3b31e5e3-72b1-4ce3-b6c2-848cf94e4128`; duration 725 s; zero scenario failures.

Table 8: Primary benchmark (GPT-4.1, 13 scenarios, job `3b31e5e3`). Ret.=Retention, Gnd.=Groundedness, Fbk.=Feedback, Bel.=Belief quality, Ovr.=Overall.

System	Ret.	Gnd.	Fbk.	Bel.	Ovr.
ACME	1.000	1.000	1.000	0.700	0.925
RAG-like	0.969	0.977	—	—	0.487
MemGPT-insp.	0.977	0.969	—	—	0.487
LangGraph-sty.	0.900	0.977	—	—	0.469

Baselines exclude feedback/belief in overall (not applicable). Scores rounded to three decimals from persisted benchmark_runs export.

Fairness of comparison. RAG-like, MemGPT-inspired, and LangGraph-style baselines do not expose explicit belief states or feedback-correction records. Their Feedback and Belief Quality cells are N/A rather than failed capabilities. The overall MemoryBench score is therefore a **system-level capability index** that rewards belief lifecycle features ACME implements—not a pure retrieval-quality

score. For retrieval-only comparison, read **Retention** and **Groundedness** separately (Table: all systems ≥ 0.90 on GPT-4.1 except LangGraph-style retention 0.900).

Key finding. ACME’s MemoryBench advantage comes primarily from **feedback correction** (1.000 vs. N/A) and **belief quality** (0.700 vs. N/A), not from higher retention or groundedness alone. On GPT-4.1, ACME retention/groundedness (1.000/1.000) are competitive with RAG-like (0.969/0.977) and MemGPT-inspired (0.977/0.969) baselines [8,9].

7.1 Model sensitivity (GPT-4.1-mini)

Job `107d02c0`, 639 s. ACME **0.858** overall vs RAG-like **0.473** (+0.39). Feedback stays at 1.000 and belief at 0.700—cognitive layers remain stable when the base model is weaker [3,34].

Table 9: Model sensitivity — GPT-4.1-mini (job `107d02c0`).

System	Ret.	Gnd.	Fbk.	Bel.	Ovr.
ACME	0.892	0.838	1.000	0.700	0.858
RAG-like	0.908	0.985	—	—	0.473
MemGPT-insp.	0.915	1.000	—	—	0.479
LangGraph-sty.	0.854	0.862	—	—	0.429

Belief quality remains **~0.70** across GPT-4.1 and GPT-4.1-mini tiers—suggesting the external belief substrate [2,20] is comparatively stable while retention/groundedness track model capability [11].

7.2 Component ablations

We disable one cognitive layer at a time via environment flags (`scripts/run_ablation_prod.sh`). Table 2 reports ACME-only MemoryBench v3.1 scores (no baseline compare — 4x cost reduction per ablation).

Full ACME row from 13-scenario compare (job `3b31e5e3`). Ablation rows from ACME-only runs on GPT-4.1 (24 June 2026, premium ingress, stamp `20260624213154`).

Key finding. Disabling belief sync collapses belief quality to **0.000** and overall score by **−0.19**, confirming the belief engine as the primary differentiator. Contrarian and vector ablations each reduce overall by only **−0.002**.

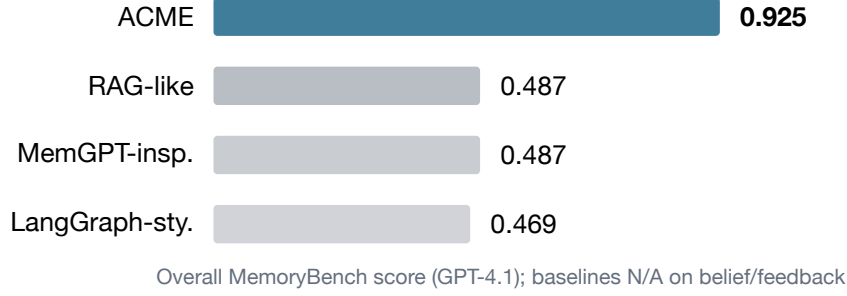


Figure 3: Result summary: ACME leads primarily on feedback and belief-quality dimensions unavailable to baselines.

Table 10: Component ablations (ACME-only, MemoryBench v3.1, GPT-4.1).

Configuration	Ovr.	Ret.	Gnd.	Fbk.	Bel.	Δ
Full ACME	0.925	1.000	1.000	1.000	0.700	—
No contrarian	0.923	1.000	0.993	1.000	0.700	−0.002
No belief sync	0.731	0.996	1.000	0.929	0.000	−0.194
No vector retrieval	0.923	0.993	1.000	1.000	0.700	−0.002

8. Limitations

Key limitations.

- LLM judge and extractor introduce variance [11,12]; we report sandbox-isolated runs with persisted benchmark_runs.
- Scenarios emphasize SaaS churn/latency plus one healthcare transfer set; broader domains and multilingual evals needed [35,36].
- Baselines are ****RAG-like / MemGPT-inspired / LangGraph-style**** reference runners in this repository, not official upstream products [9,10].
- MemoryBench is author-defined; external human rating of a scenario sample is future work.
- Deployed API requires an API key for benchmark endpoints.
- CRS weights are hand-tuned; calibration against human epistemic judgments [39] is future work.
- **LongMemEval routing.** The LongMemEval adapter uses per-type answer paths (transcript-first for KU, vector-ranked aggregation for multi-session, abstention and preference prompts); MemoryBench still uses the graph + CRS query path.
- **Multi-session parity.** After v4 routing, ACME matches the RAG-like baseline on multi-session (79.3% each on the 241-Q v4

run); dense retrieval remains competitive when evidence is scattered without temporal conflict.

- **Abstention.** Full-split abstention accuracy reaches 83.3% (v5) via anchor short-circuit; harder entity-substitution probes remain imperfect.

9. Conclusion

ACME externalizes belief lifecycle, contradiction handling, and feedback-driven correction outside LLM weights. On MemoryBench v3.1, gains concentrate on **feedback** and **belief quality** dimensions that RAG-like and MemGPT-inspired baselines cannot report; retention and groundedness remain competitive. On LongMemEval (500 Q, GPT-4.1), hybrid transcript routing reaches **87.6%** vs. **77.6%** (RAG-like), with the largest margins on knowledge-update and temporal-reasoning types. We do not claim universal SOTA on every long-context benchmark; rather, the results support a narrower hypothesis: **explicit external belief management complements retrieval and long-context modeling** for long-horizon agents [20,21,48]. Concurrent systems [48,52,53] pursue similar directions; ACME contributes an open, sandbox-isolated eval-

uation protocol and persisted benchmark_runs for third-party audit [47].

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11. Appendix A: Reproduction

```
git clone https://github.com/KamilBourouiba/ACME.git && cd ACME
pip install -e ".[dev]"
pytest tests/test_benchmark_gate.py tests/test_memorybench.py -q
# Production (requires API key):
./azure/configure-premium-ingress.sh
./scripts/run_prod_benchmark.sh
./scripts/run_model_benchmark.sh gpt-4.1-mini # optional model sweep
./scripts/run_ablation_prod.sh # component ablations
```

12. Appendix B: Data Availability

Benchmark payloads are stored in PostgreSQL table `benchmark_runs` (columns: `overall_score`, `retention_score`, `belief_quality_score`, `payload` JSONB, `created_at`). Export via GET `/api/v1/benchmark/export`. Full run tables are documented in `docs/BENCHMARK_RESULTS.md`~[4

13. Appendix C: Preview model sensitivity (GPT-5.4)

Job `eb68f028` (620 s) on an Azure OpenAI preview deployment (`gpt-5.4`, `TESTING-CHAT-BETA` resource). This model name was available on our tenant at benchmark time but is **not** listed in OpenAI’s public GPT-4.1 documentation [45]; treat as supplementary.

Table 11: Model sensitivity — GPT-5.4 (job `eb68f028`).

System	Ret.	Gnd.	Fbk.	Bel.	Ovr.
ACME	0.952	0.839	1.000	0.701	0.873
RAG-like	0.964	0.931	—	—	0.474
MemGPT-insp.	0.975	0.939	—	—	0.478
LangGraph-sty.	0.973	0.950	—	—	0.481

ACME **0.873** overall vs RAG-like **0.474** (+0.40); belief quality **0.701**. Pattern matches GPT-4.1-mini sensitivity: cognitive layers stable, overall gap driven by N/A baseline belief/feedback columns.